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Design and Interaction Interface using Augmented Reality for Smart Manufacturing

Yunbo Zhang[a], Tsz-Ho Kwok[b,*]

[a] Department of Mechanical Engineering, Tennessee Technological University, Cookeville, Tennessee 38505, USA
[b] Department of Mechanical, Industrial and Aerospace Engineering, Concordia University, Montreal, QC H3G1M8, Canada

Abstract

In this paper, we apply Augmented Reality (AR) technologies to develop a design and interaction interface for Smart Manufacturing (SmartMFG). This work is motivated by the lack of appropriate human-machine-interaction (HMI) tools to support interaction and customization in SmartMFG environment. Trying to address this research problem, we hypothesize that AR-based design interfaces that communicate with Machine Control Unit (MCU) directly will increase the degree of interaction and the complexity of instructions performed in Manual Data Input (MDI) systems. To test this hypothesis, we developed a prototyping system consisting of an AR-tablet device as the input interface and an Ultimaker 3 printer as the machine tool. Firstly, this AR-based system has sensing, design and control capabilities to interact and communicate with the machine tool via Wifi. Secondly, a set of sketch-based computational tools is developed for users to design shapes on existing objects easily and efficiently within the AR environment. Finally, The customized design is converted to machine code, which is also customized based on the machine tool and the registration of the virtual model and the existing object. We tested our system by designing two customized shapes onto an existing shape in the AR environment and generating the G-code to control the printer to fabricate them onto the physical object.

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Keywords: computer-aided design, smart manufacturing, industry 4.0, augmented reality, additive manufacturing, 3D printing

1. Introduction

Smart Manufacturing (SmartMFG), also known as Industry 4.0, is the new trend of automation and data exchange in manufacturing technologies. According to National Institute of Standards and Technology (NIST), SmartMFG are "fully-integrated, collaborative manufacturing systems that respond in real time to meet changing demands and conditions in the factory, in the supply network, and in customer needs" [1]. As the next phase in the digitization of the manufacturing sector, the foundation of SmartMFG consists of Cyber-Physical Systems (CPS) and Internet of Things (IoT), and it aims to connect data seamlessly across different stages of the manufacturing process with latest sensing technologies and wireless connections. Together with the advance of three-dimensional (3D) printing processes, it is possible to produce one-of-a-kind product without increase in cost and lead time.

In the Distributed Numerical Control (DNC) systems, each machine tool, e.g., milling and 3D printer, is connected to a machine control unit (MCU), e.g., computer. The MCU controls the machine tool by sending part program, which is a set of instructions to be performed by the machine. Part programming can be accomplished using a variety of procedures ranging from highly manual to highly automated methods: manual part programming, computer-aided design and manufacturing (CAD/CAM) part programming, and manual data input [2]. In manual part programming, the programmer prepares the codes using a low-level machine language, e.g., G-code. A CAM/CAM system is a computer interactive graphics system that can define part

* Corresponding author. Tel.: +1 (514) 848-2424 #3807.
E-mail address: tszho.kwok@concordia.ca

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Fig. 1. An AR interface is adopted to interact with manufacturing systems.

geometry and generate tool path to create the machine-language part program. It automates a substantial portion of the procedure, but a significant commitment in equipment, software, and training is required. To make our factory smart, one way is to have the machine operator perform the part-programming task at the machine tool, which is called manual data input (MDI). A minimum of training in part programming is required of the machine operator. However, the current development requires the operator to enter the part geometry data and motion commands manually into the MCU, and thus MDI can only handle simple operations and parts. There is a critical need to have appropriate tools of human-machine interaction (HMI) to support interaction and customization in SmartMFG environment. The research question is: *what should the HMI system look like such that individuals can access and interact within SmartMFG to design and manufacture customized products?*

With the growth of Augmented Reality (AR) technologies, AR devices can be employed to help increasing safety in production plants and reducing physical demand to workers (Fig. 1). The attractive feature of AR is that it brings the information of physical environment to inspire, contextualize, and guide the users' creativity. AR techniques have been explored in various applications, including virtual content creation [3–7], assembly guidance [8–11], control interface of robots [12–15], and simulation of manufacturing processes [16,17]. There are also researches [7,18] introducing AR-based design systems to create virtual designs around/on existing objects. Designing 3D shapes around a given reference physical objects in AR environment provides the shape and dimensionality of the existing object to design customized products based it. Moreover, the instant visual feedback enabled by AR also helps users to better evaluate their designs and modify them accordingly. Trying to answer the research question, this paper hypothesizes that **AR-based design interfaces that communicate with MCU directly will increase the degree of interaction and the complexity of instructions performed in MDI systems.**

To test this hypothesis, we developed a prototyping system consisting of an AR tablet device and an Ultimaker3 3D printer as the machine tool. Based on an Asus Zenfone smart phone powered by Google Tango AR toolkit, our software system leverages the depth images of the physical object and allows users to interact virtually with the build platform. Users can sketch 2D curves on the existing object and convert the 2D curves into 3D shapes via simple interactions. After the designs are created, our software system will convert them into a set of instructions that controls and directs the machine tool. The instructions are sent directly to the 3D printer with the wireless network, so that the newly added 3D shapes will be printed onto the existing objects. The contribution of this paper is summarized as:

1. An AR-based design and manufacturing environment is proposed with sensing, design and control capabilities to interact and communicate with the machine tool via Wifi.
2. A set of sketch-based computational tools is developed for users to design shapes on existing objects easily and efficiently within the AR environment.
3. The customized design is converted to machine code, which is also customized based on the machine tool and the registration of the virtual model and the existing object.

In addition to the 3D printing application, this system can customize and fabricate products based on some existing objects, instead of always 3D-printing product from scratch.

2. Related work

We classify the related works broadly into AR-based virtual content creation, and AR-based assembly, control, and process simulation.

2.1. AR-based Virtual Content Creation

Researchers have explored virtual content creation using an AR-based system. A previous work [3] integrated tangible instruments for modifying virtual models. Nuernberger et al. [4] interpreted 2D gesture annotations in 3D AR environment. SnapToReality [19] extracted 3D edges and planes from the environment

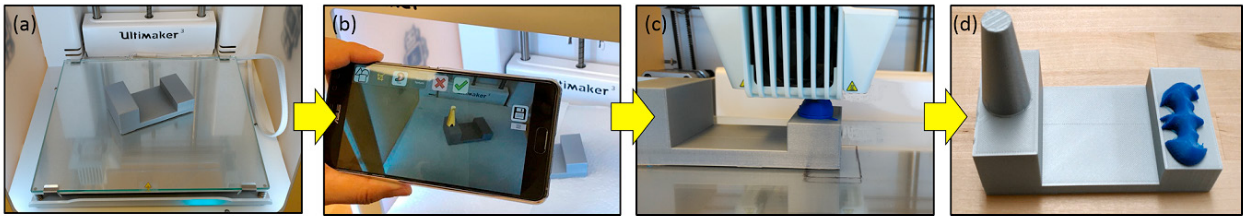

Fig. 2. The workflow of the proposed system: (a) The basic object is placed in the Ultimaker3 3D printer, and (b) users can create and edit new shapes onto the existing object through a Zenphone AR device. (c) After a few step of creation and editing, users obtained the 3D design in the AR environment, and pass them to 3D printer. (d) Eventually, customized 3D shapes are added physically on the existing objects.

and utilized them for aligning 3D objects precisely. Lau et al. [5] attached fiducial markers onto physical primitives and assembled corresponding virtual models to the physical primitives. GARDE [6] presented gesture based AR design systems for evaluation designs.

A recent work – MixFab [7] – allowed users to integrate scanned 3D objects to design using mid-air gestures in an AR desk environment. Window-Shaping [18] provided a sketch interface to directly design 3D shapes on physical objects. Inspired by Window-Shaping, our work adopts a sketch interface to create 3D designs on existing objects for human-machine interaction by integrating AR technologies in the MDI system.

2.2. AR-based Assembly, Control and Simulation

AR has been widely explored to assist users in assembly process by providing virtual guidance. Virtual instructions were overlaid in the view of user's AR headset [8], while corresponding virtual CAD models were displayed around physical objects [9]. Different interaction metaphors were adopted in the AR-based assembly process, including a virtual interactive tool [10] or bare hand for assembly guidance [11]. By integrating *Radio-frequency identification* (RFID) into each of the construction modules, A real-time tracking and monitoring was enabled for assembly guidance [20]. Yet, these work focused on providing assembly guidance, not 3D design creation.

Moreover, AR has been investigated for interacting with robots. TouchMe [12] adopted AR to control the mobile robots. Leveraging AR with robots, researchers have achieved human-robot collaboration for object manipulation [13], object pick and place [14], and household sequential task instruction [15]. The control interface of the interaction with robots could be either an AR tablet device or a projected interface on or near the physical targets [21] for ease of interaction. In industrial applications, AR has been applied in industrial robots for path planning [22,23], spatial programming [24], and trajectory planning [25]. AR has also been used for simulation of manufacturing process. Different manufacturing processes, including metal casting [16] and CNC machining [17], have been simulated in AR environment for verifying process plan, reducing material cost, and training novice mechanist.

However, none of these above works focuses on designing and manufacturing customized shapes on an existing object. We aim at investigating an unified AR framework allowing users to design customized shapes and manufacture them on-the-fly.

3. Overview

Our AR environment has three main modules to support design and manufacturing of customized products, including 3D sensing of physical environment and objects, Wifi-based communication and control, and in-situ design system for customized shape. The depth camera on Zenphone device allows the acquisition of 3D point cloud data of the physical environment. These point cloud data provide the information of the shapes of the existing objects, and are the basis for our design system. To manufacture the designed 3D shapes, the AR device are connected to manufacturing systems via Wifi. All the manufacturing planning are also done on the AR device and the part programs are directly passed to the machine control unit. The data of the design and manufacturing across the AR device and machine are transferred by the communication module. The proposed AR environment also supports design of customized 3D shapes with respect to the existing objects. Leveraging the sensed 3D information of the existing objects, the in-situ design system enables users to design customized products on/around existing objects. The AR environment also provides new affordances for design such as touch interactions.

We implement a prototype system to prove the concept of the proposed AR environment. This prototype system consists of an a Zenphone AR 5.7 inch smart phone [26] with Qualcomm Snapdragon 821 processor and 8GB RAM, running Android 7.0 KitKat OS, and an Ultimaker 3 Fused Deposition Modeling (FDM) 3D printer with a Wifi communication module on it [27]. The reason we pick Ultimaker 3 as our machine tool is that its software and firmware open-source. The manufacturer of Ultimaker discloses all the source code and technique specification of their products, which allows developers to study its infrastructure and tailor their application with the machine. The software system is developed using C++ in Android Studio 2.3.3 environment with Android NDK (JNI) for 3D shape modeling, and OpenGL shading language for rendering 3D objects. The Tangle SDK [28] utilized in the software system allows us to access the 3D point cloud of the physical scene. The communication between the Zenphone and the Ultimaker 3 is established on the support of Ultimaker 3 developer API [29].

Currently, most 3D printers create the 3D objects from scratch. The manufactured 3D objects after printing is fixed and can not be modified. Our prototype system enables the editing of existing physical objects by allowing users to design customized shapes on the existing objects and physically add them onto the existing object using 3D printer. As shown in Fig. 2, the workflow of the proposed prototyping system unfolds as follows: the user places an exiting object into the 3D printer (Fig. 2(a)), creates and designs the customized 3D shapes onto the existing object using our sketch based design interfaces (Fig. 2(b)), and sends the designed 3D shapes to 3D pinter for adding them physically onto the existing object (Fig. 2(c,d)). The Ultimaker 3 printer is connected with the AR device via Wifi connection. To make sure the designed 3D shapes properly to be printed onto the right position on the existing object, the coordinates system of AR devices and the physical coordinates system of 3D printer are aligned by a calibration process. Our software system customized the G-code such that the designed shaped can be added onto the surface of the existing object instead of the original 3D printer's platform.

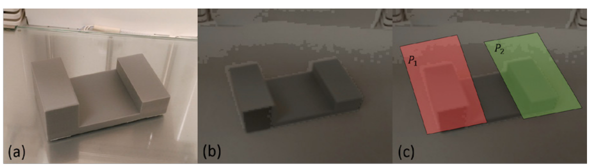

Fig. 3. The depth camera on the Zenphone AR device provides the depth sensing of physical environment: An existing base shape (a) is sensed by the depth camera on the Zenphone; the acquired depth image (b) is used to fit planes (c) for the following modeling procedure.

4. MDI System

Our system consists of three modules: 1) a 3D sensing module, 2) Wifi-based communication module, and 3) an in-situ design module. The detailed description is presented as follows.

4.1. 3D Sensing for Physical Environment

The Zenphone AR device with a depth camera enables the depth sensing of the physical environment. The depth camera on Zenphone AR device captures depth data (5Hz) of the physical environment around. As shown in Fig.3, an existing physical object (Fig.3 (a)) is sensed by the depth camera, and the depth data is acquired and stored (Fig.3 (a)). The obtained point cloud provides the 3D geometric information of the physical environment and the objects in the environment. Specifically, the built-in Google Tango toolkits fit a set of planes to the obtained point cloud (Fig.3 (c)). These planes serve as the basis to design 3D shapes around the existing objects. Our design system with a set of geometric modeling and interaction tools is actually built upon the fitted planes.

4.2. Wifi-based Communication between Devices

In our prototype system, Zenphone AR device communicates with Ultimaker 3 3D printer via a wifi connection. The Ultimaker 3 printer particularly fits our needs because that it has a new communication module and allows users to access it through Wifi. After connecting to the lab's Wifi network, Ultimaker 3 printer creates a Wifi hotspot, and an IP address is assigned to the printer. Ultimaker 3 allows to be directly accessed by the 3rd party software using the assigned IP address

Fig. 4. The Asus Zenphone AR smart phone communicates with the Ultimaker 3 printer through the Wifi hotspot created by the communication module of the printer.

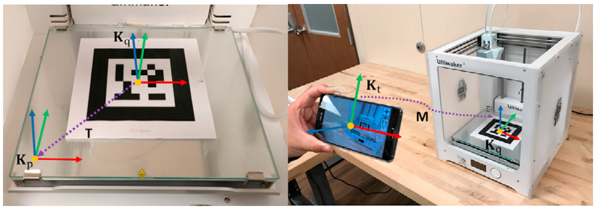

Fig. 5. A Quick Response (QR) code based 3D positioning method is conducted to align the coordinates system of the AR environment $\mathbf{K}_t$ and the one of the physical 3D printer $\mathbf{K}_p$ in 2 steps: 1) a printed QR code is carefully placed onto the platform of the 3D printer such that the translation vector $\mathbf{T}$ between the coordinates system $\mathbf{K}_q$ located at the center of the QR code and $\mathbf{K}_p$ is measured (left); then the 3D positioning is performed and the transformation matrix $\mathbf{M}$ is computed (right). With $\mathbf{M}$ and $\mathbf{T}$, $\mathbf{K}_t$ and $\mathbf{K}_p$ can be aligned.

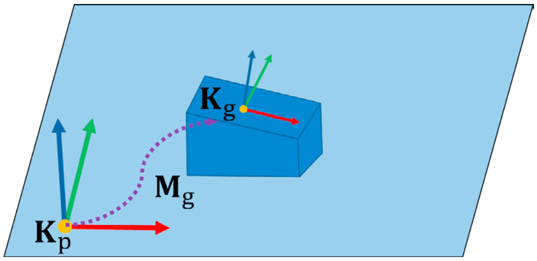

Fig. 6. The transformation matrix $\mathbf{M}_g$ between $\mathbf{K}_p$ and the local coordinates system $\mathbf{K}_g$ is computed for generating our customized G-code.

and preset password in "Developer Mode". Users can control the printer using the built-in API functions of Ultimaker 3. Specifically, in our prototype system, API function "sendgcode" allows us to send customized G-code to Ultimaker 3 printer. For detailed description of the API, please refer to Ultimaker 3's document [29].

4.3. Coordinates Systems Alignment

Our system allows users to create customized shapes on the existing physical object in the AR environment. Then, these created shapes will be printed onto the existing object. It requires a registration to align the coordinates system of the AR environment $\mathbf{K}_t$ to the one of 3D printer $\mathbf{K}_p$ in physical world.

To find this registration, we conducted a Quick Response (QR) code based 3D positioning, which consists of 2 steps: 1) a printed QR code is attached on to the printing platform of the printer. According to Ultimaker 3's developer manual [29], the origin of $\mathbf{K}_p$ is located at the lower-left corner of the printer's platform and the 3 axises are spanned like the ones shown in Fig.5 (a). Another coordinates system $\mathbf{K}_q$ with its origin located at the center of the QR code is also established, and the translation vector $\mathbf{T}$ between $\mathbf{K}_q$ and $\mathbf{K}_p$ can be measured. 2) Then, the 3D positioning method is conducted to find out the transformation matrix $\mathbf{M}$ between $\mathbf{K}_t$ and $\mathbf{K}_q$. We utilized the built-in function "$TangoSupport\_detectMarkers$" in Google Tango Toolkits [28] to conduct the 3D positioning with QR code. With $\mathbf{M}$ and $\mathbf{T}$ computed, any 3D position $\mathbf{v}$ on the designed 3D shape in $\mathbf{K}_t$ can be transformed to a corresponding position $\mathbf{v}'$ in $\mathbf{K}_p$ via: $\mathbf{v}' = \mathbf{M}\mathbf{v} + \mathbf{T}$. As the origin of $\mathbf{K}_t$ and its 3 axes are set up to be the centroid and local axises of the device when the application is run on it [28], this coordinates alignment needs to be perform once when the our software system is started.

4.4. Customized Slicing & G-code Generation

The original G-code generated for Ultimaker 3 assumes the printed objects laying on its original platform. In our prototype system, the customized 3D shapes are designed on some fitted planes on the existing objects. Therefore, the original G-code generated by the standard G-code generator can not be used directly. We need to customized the G-code for the designed objects. Specifically, with a designed object S, we need to transform it with respect to the 3D printer's physical coordinates system $\mathbf{K}_p$. Then, we establish a local coordinates system $\mathbf{K}_g$ on a fitted plane (shown in Fig.6). $\mathbf{K}_g$'s local Z-axis is aligned with the normal of the plane, and its local origin point is set to be the average position of the input points laying on the plane. The local X-axis can be created by picking a random points on the plane and forming a vector from the local origin to the chosen point on the plane. After that, local Y-axis can be obtained by a cross product of local X-axis and local Z-axis. The transformation matrix $\mathbf{M}_g$ between $\mathbf{K}_p$ and $\mathbf{K}_g$, and its inverse matrix $\mathbf{M}_g^{-1}$ can be computed by forming a Jacobi matrix and conducing a Singular Value Decomposition (SVD) on the matrix [30]. The object S is transformed into local coordinates systems using $\mathbf{M}_g$, and then sliced using a standard slicer. These slices can be transformed back into $\mathbf{K}_p$ coordinates system by a inverse transformation with $\mathbf{M}_g^{-1}$. Afterwards, the customized G-code can be generated based on the transformed slices.

5. AR-based Design System

To develop a system enabling the design of customized products in AR environment, a challenging question is: what kind of metaphors of interactions and schemes of modeling should be adopted in such a system? For conventional CAD systems, it is quite difficult to design customized products around some reference shapes. As these system base interactions on Windows-Icons-Menus-pointer (WIMP), it is very time consuming to fit a design to an existing 3D shape, and requires large numbers of translation and rotation operations to change the viewpoints in the design interfaces. Moreover, existing CAD tools are complex and hard to learn, which prevents the adoption of CAD from broader user base, i.e. novice users.

Different from conventional CAD tools, we built up a sketch-based design system following the design system in [18] to support customized shape design. Leveraging the 3D point cloud data acquired, our system allows users to design customized shapes on/around the existing object. The use of Zenphone device provides the capabilities for both spatial and multi-touch interactions. The 3D mobility of the Zenphone device allows user to easily adjust the viewpoint of the design interface to get feedback of the created shapes with respect to the existing object. On the other hand, the touch screen enables both precise and intuitive shape creation by users' touch and 2D sketch. We adopted the 2D curve as the basic representation of shapes, as it is easy to create, edit, and be used as the basis for creating 3D shapes [31]. We developed a set of interactive tools to create, edit and manipulate 2D curves, as well as a inflation-based modeling tools to create 3D shapes based on the 2D curves.

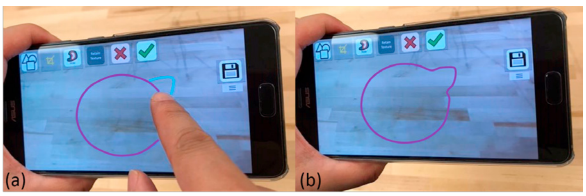

Fig. 7. The sketched curve (a) can be continuously edited by users using the over-sketch editing tool (b).

5.1. Interactive Modeling & Editing Tools

2D Curve Sketching and Editing. Our system allows users to sketch 2D curves on the Zenphone's touch screen. Once the curve is sketched, it is projected onto a fitted plane based on the 3D point cloud acquired from the existing objects in physical scene. The sketched curve can be modified continuously by over-sketching near the curve until a satisfactory shape is obtained (shown in Fig.7). As the sketched curves will be used as the boundary curves to generate 3D surfaces, they are required to be closed and smoothing. To meet these requirement, we applied an exponential smoothing [32] and then an equidistant curve re-sampling [33] to the

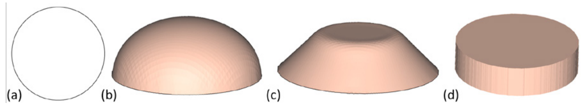

Fig. 8. The input sketch (a) can be inflated into different shapes according to different inflation functions [35], including the Circular (b), conical (c), and linear (d) functions.

sketched curves. The work flow of curve sketching and editing id described as follows: users sketch 2D curves on the touch screen of Zenphone by either their fingers or stylus. The original sketched curves formed by sequences of points, and these points are neither evenly distributed nor smooth. Therefore, an exponential smoothing method is applied to the sketch dynamically. During user's sketch, the processed points can be obtained via following formula:

$$\mathbf{s}_t = \alpha \mathbf{x}_t + (1 - \alpha)\mathbf{s}_{t-1}, \quad (1)$$

where $\mathbf{s}_t$ is the processed point at time current t, $\mathbf{x}_t$ is the original point at time current $t - 1$, $\mathbf{s}_{t-1}$ is the processed points at time current $t-1$, and α is the smoothing factor chosen between 0 and 1 set by users. To start, $\mathbf{s}_0$ is set as the first input point $\mathbf{x}_0$. After the smoothing process, a equidistant curve re-sampling is applied to generate points evenly distributed on the curves. To edit the local shape of the sketched curve, users can over-sketch some new shapes near the existing. The newly added curve will change the shape of existing curve by replacing the nearby portion of the exiting curve.

Applying Template Curve. An alternative way to create 2D curves is to place a pre-defined template curve onto the existing object. The curve templates are either created by sketching or by extract contours of objects in an image. We implemented a contour extraction tool using GrabCut algorithm [34]. GrabCut algorithm is developed based on optimization by graph-cut, which extracts the contour of a certain object from an input image by simple interaction. The user can import an image they find from Internet, and interactively select the object in the image. Then, the GrabCut algorithm will automatically generate the contour of the selected object. This contour curve can be stored as a template curve, and it can be imported into our design system. Once the template curve is selected by the user, it can be placed onto any plane in the scene by user's single-tap gesture.

Inflation based Surface Modeling. Once the 2D curve is finalized, an initial 2D mesh surface is generated using Constrained Delaunay Triangulation (CDT) [36].

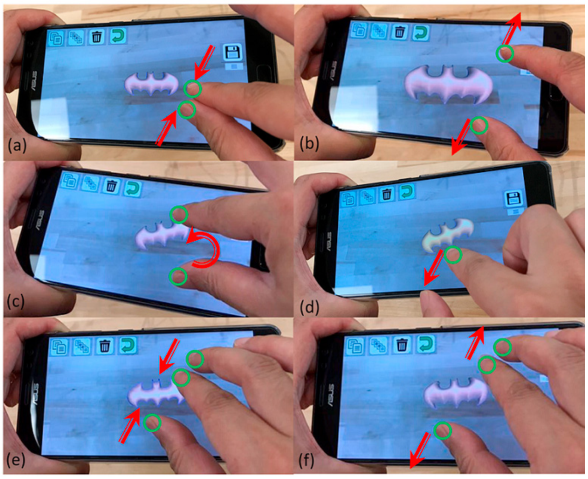

Fig. 9. The two-finger touch pinch/spread operation scales down (a) or up (b) the created 3D shapes. The 3D shape can be rotated in the same plane of its boundary curve using the two-finger rotation gesture (c). The translation of the 3D shape is realized using one-finger touch and move (d). The three-finger pinch/spread gestures enable the modeling control on the inflation tool by pulling the created shapes out of (e) or pushing into (f) the screen.

Then a inflation operation is applied onto the generated 2D mesh surface to generate a 3D shape. On the initial 2D mesh surface, an inflation function X_d is defined at each vertex on the surface, where d is the distance between a vertex and the boundary curve. X_d determines how each vertex on the surface is inflated. Following [35], different X_ds (circular, conical and linear) are adopted to enable different modeling capability (shown in Fig.8). This inflation based modeling tool allows the individual users to create 3D shapes easily and efficiently.

5.2. *Gesture based Interaction*

Our design system leverages the 2D touch gestures of smart phone to enable intuitive user interactions including shape manipulation and modeling control.

Shape Manipulation. The basic shape manipulations such as rotation, scaling are implemented on the Zenphone device using two-finger touch gestures (Fig.9 (a)-(c)). Specifically, the selected 3D object is rotated in the same plane of its boundary curve when two fingers rotates on the touch screen of the Zenphone (Fig.9 (c)), while the scaling is realized by pinch/spread the two fingers (Fig.9 (a) and (b)). The translation of the created 3D shapes is achieved by one-figure touch and move (Fig.9 (d)).

Modeling Control. In the inflation operation for the shape modeling, users are able to control how much the 3D shapes are inflated by a three-finger gesture. Here, pinching three fingers pulls the created 3D shape out of the screen (see Fig.9 (f)), while spreading results in pushing the shape into the screen (see Fig.9 (e)).

6. Results and Discussion

We tested our system by designing two customized shapes onto an existing block-like object (see Fig.2). For the existing object laying on the Ultimaler 3 printer's platform (see Fig.3 (a)), the depth sensor on Zenphone device successfully acquired the 3D point cloud data of the object (Fig.3 (b)). These point cloud data are then fitted by a set of plane as the basis of the following design procedure.

To test the sketch based design tools, we sketched a circle onto one fitted plane of the basic shape. The 2D touch gesture based interactive tools were used to edit the 2D sketched curve and control the modeling method based on inflation. The conical distance function [35] for inflation was also tested to generate 3D shapes with sharp features, and the cone-like shape was created (Fig.2). The inflation-based modeling method enabled users to create the 3D shape easily and efficiently. Another test on the template curve function was performed. We chose an image of the batman logo among a set of template images (Fig.10(a)), and extracted the outer contour as a curve (Fig.10(b) middle). This template curve then was inflated to a 3D shape using circular distance function (Fig.10(b) bottom). The template curve function enables users to repurpose the images from other resources, and enhances the curve design capability of our system.

The generated 3D designs were then converted into customized G-codes, and they were sent to the 3D printer to fabricate 3D shapes onto the existing object (Fig.2 (d)). The Wifi-based communication module of our system supported the data transfer between AR device and 3D printer. The 3D printer was controlled by the G-code sent from the AR device. Another essential functionality was tested is the coordinates alignment and customized G-code generation. The success of printed results shows that the 3D positioning based on QR-code can aligned the physical coordinates system of the 3D printer and the virtual coordinate system in AR environment, and the customized process planning can successfully generate appropriate G-code for manufacturing.

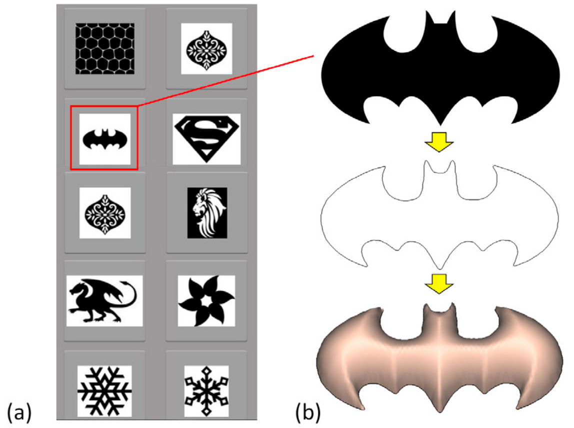

Fig. 10. From a set of pre-defined template images (a), an image of Batman logo is chosen. Then the contour of this image is extracted ((b) middle), and the contour curve can be used to generate a 3D shape via inflation operation ((b) bottom).

Although our current system is only a prototype system, it successfully demonstrated the AR-based MDI system design. This system realized the efficient and intuitive interactions between users and machine tools in smart manufacturing system. The three main modules of the system including sensing, design and control modules were verified to effectively and efficiently support the design and production of customized shapes.

7. Conclusion and Future Work

In this paper, we proposed an AR-based design and manufacturing environment for smart manufacturing systems to enable individual users to produce customized products. This AR environment allows users to design customized products and manufacture them by accessing, monitoring and controlling the connected manufacturing systems. We implemented a prototype systems consisting of a Zenphone AR smart phone and an Ultimaker 3 printer. We established the communication of the Zenphone device with Ultimaker 3 printer through the Wifi network. The physical coordinates system and the AR environment's coordinates system are aligned by a calibration based on QR code. We also implemented an AR based design system allowing user to design customized shapes onto the existing objects. The design system leveraging the 3D point cloud data acquired by Zenphone device enables users to sketch and edit 2D curves, and projects the curves onto the fitted planes. The developed touch-based interactive tools provides the intuitive and easy interaction for users. Our geometric modeling method creates 3D shapes by an inflation operations. The created 3D shapes are aligned to the physical coordinates system of the 3D printer and the G-codes are generated and sent to the printer. We tested our system by printing two customized shapes onto a basic object as shown in Fig.2.

As our current system is a prototype system with limited modeling capability, we plan to enhance the modeling capability for more complex design. Although we assume that our current system with touch-based interaction is easy-to-use and intuitive for individual users, it still requires comprehensive user study to guide the interaction design for an AR based modeling system. Currently, we build prototype system based upon an Ultimaker3 3D printer. However, the modeling capability of the proposed design system should not be restricted by 3D printing. In the future, we will enhance our system by introducing more complicated curve-based modeling to support other manufacturing processing such as CNC machine and laser cutting. Besides design, we will also to extend the functionality of our system to manufacturing process monitoring and repairing to further leverage AR technique's advantages. We believe that AR-based design and manufacturing environment will greatly support the accessibility of smart manufacturing.

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